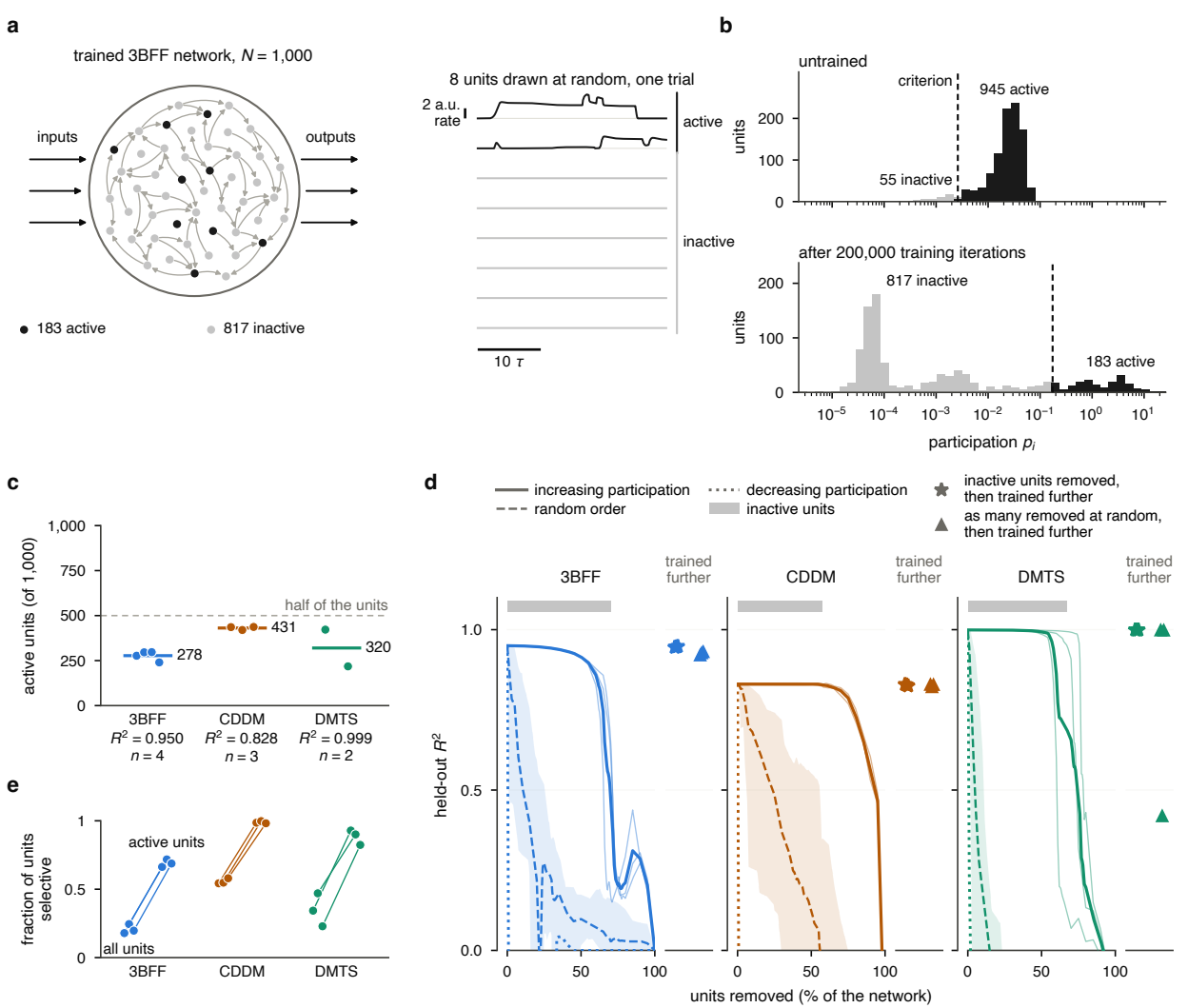

a
trained 3BFF network, N = 1,000
inputs
outputs
183 active
817 inactive
8 units drawn at random, one trial
2 a.u.
rate
active
inactive
10 τ
b
untrained
criterion
945 active
55 inactive
units
200
100
0
after 200,000 training iterations
817 inactive
183 active
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1}
participation p_i
c
active units (of 1,000)
1,000
750
500
250
0
half of the units
431
278
320
3BFF
$R^2 = 0.950$
$n = 4$
CDDM
$R^2 = 0.828$
$n = 3$
DMTS
$R^2 = 0.999$
$n = 2$
d
increasing participation
random order
decreasing participation
inactive units
inactive units removed, then trained further
as many removed at random, then trained further
3BFF
trained further
CDDM
trained further
DMTS
trained further
held-out R^2
1.0
0.5
0.0
0
50
100
units removed (% of the network)
e
fraction of units selective
1
0.5
0
active units
all units
3BFF
CDDM
DMTS